

Chapter

Internationalization and Firm Performance of Firms in India: The Role of Top Management

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Abstract

This study analyzes the relationship between the degree of internationalization and firm performance of 380 firms in the manufacturing and service sectors in India, considering the moderating effects of top manager characteristics—specifically gender and managerial experience. Using Feasible Generalized Least Squares (FGLS) regression on panel data from the World Bank Enterprise Surveys, results reveal: (1) internationalization negatively impacts firm performance initially due to high entry costs, including market unfamiliarity, cultural barriers, and resource constraints; (2) top managers' experience and gender diversity (female leadership) positively moderate this relationship, enhancing performance by fostering innovation, adaptability, and strategic responsiveness. The research contributes to internationalization theory by extending the Uppsala model and Upper Echelons Theory through the integration of managerial characteristics, providing valuable insights for managerial practice and policy in developing countries.

Keywords: internationalization, firm performance, top management, gender diversity, managerial experience, Uppsala model, Upper Echelons Theory, India

1. Introduction

In the field of International Business, “Internationalization” is the strategy that businesses implement to expand their operations overseas to increase revenue. Many studies have focused on the relationship between internationalization and firm performance; however, the empirical results are inconsistent, showing contradictions and even conflicts [1–4]. The relationship between internationalization and firm performance has yielded mixed empirical findings [3, 4]. Recent research highlights the significance of moderating variables such as top management characteristics in clarifying this relationship [5, 6]. Employing the Uppsala model [7, 8] and Upper Echelons Theory [9], this study investigates how managerial gender and experience moderate the internationalization-performance link in Indian firms.

Since 2007, studies have integrated moderating factors into the model and found that these factors can positively or negatively impact the relationship between internationalization and business performance [5, 6]. Ignoring these factors can

lead to biased predictions, especially when top management characteristics, such as age, tenure, international experience, and education [10], strongly influence the internationalization process. Top managers with experience and flexibility in decision-making help businesses adapt better to the global environment. Moreover, the internationalization process is considered an important strategy to help businesses expand their markets and increase revenue [11], but the empirical results on the relationship between internationalization and firm performance remain inconsistent. Anderson and McKenzie [12] emphasized that the local economic context and top management characteristics are important factors affecting the effectiveness of internationalization, especially in developing economies. Recent studies have also pointed out the role of moderating factors, such as top management characteristics, in clarifying this relationship [6]. Therefore, more research is needed on the role of top management characteristics in the context of developing economies to better understand how they influence internationalization strategy and firm performance. This study aims to answer the question: “Do top management characteristics directly influence firm performance through decision-making ability?”

This study will make three main contributions to the theoretical basis of internationalization: (1) examining the relationship between the degree of internationalization and firm performance in Indian firms; (2) using the Uppsala model and Upper Echelons Theory to analyze the moderating role of top management characteristics; and (3) providing empirical evidence on the impact of internationalization on firm performance in the context of developing economies.

2. Literature review

2.1 Theoretical background

In internationalization research, various theories explain the benefits and costs of internationalization and its impact on firm performance. These include Transaction Cost Theory, the resource-based view, agency theory, the Uppsala model, behavioral theory of the firm, and the eclectic paradigm. This study applies Johanson and Vahlne’s Uppsala model, both the original (1977) [7] and revised versions (2017, 2020) [13, 14], incorporating Hambrick and Mason’s [9] Upper Echelons Theory for support. This approach explains the moderating role of top management team (TMT) individual characteristics in the relationship between internationalization and firm performance.

2.2 Uppsala model

In this study, the research model is built upon the 1977 Uppsala model [7] to explain the involvement of businesses in the international market. According to this model, internationalization is a crucial growth strategy, progressing through four stages: (1) no regular export, (2) export through independent representatives, (3) establishment of sales branches abroad, and (4) production in foreign markets [7]. This stage-wise progression allows firms to gradually build experiential knowledge and reduce uncertainty when entering foreign markets. The model emphasizes that the knowledge, understanding, and international experience of managers play a central role in shaping the pace and scope of internationalization [7].

The 2017 and 2020 revised versions of the model [13, 14] further elaborate on the dynamic nature of internationalization by introducing four key elements: capabilities,

commitment processes, knowledge development, and relational networks. These additions reflect the shift toward a network-based view of international expansion in uncertain global environments. In such settings, top managers must demonstrate strategic judgment, adapt value chains, and manage cross-border relationships to secure long-term growth [13].

Furthermore, both the original and revised versions of the Uppsala model underscore the importance of managerial experience in reducing liability of foreignness, especially in culturally or institutionally distant markets. However, some scholars caution that internationalization may incur high costs and risk for firms lacking sufficient managerial resources or international embeddedness [15].

2.3 Upper echelons

Upper Echelons Theory refers to individuals with high social standing in a business, such as the CEO, top managers, or the top management team [16]. These individuals have their own cognitive backgrounds and values, and they evaluate business situations based on their personal perspectives [17]. This personal perspective influences the perception and strategic business decisions of the enterprise [16].

This theory, initiated by Hambrick and Mason [9], emphasizes that a company's strategic decisions reflect the personal characteristics of its managers. Characteristics such as gender, experience, and psychological traits of the manager play a crucial role in shaping the company's internationalization strategy. Empirical studies [18] have confirmed the significant role of gender and experience in creating substantial differences in how businesses respond to internationalization challenges. Additionally, Hambrick and Mason's [9] theory adds that psychological characteristics and the top manager's experience not only influence internationalization strategy but also shape how businesses seize opportunities and cope with risks in the global business environment.

This study draws from Upper Echelons Theory to emphasize that top managers' gender and experience serve as cognitive frames that shape how they interpret internationalization risks and opportunities. Female executives often bring relational skills and cultural sensitivity, supporting adaptability in diverse global markets [19]. Experienced managers, by contrast, leverage accumulated knowledge to mitigate uncertainty and reduce liability of foreignness [20].

2.4 Hypothesis development

2.4.1 The relationship between the degree of internationalization (DOI) and firm performance (P) of firms in India

The relationship between internationalization and firm performance is central to international business research. Most empirical studies suggest a positive linear relationship between internationalization and firm performance; as firms expand internationally, their performance tends to improve. However, some studies have also found nonlinear patterns, such as U-shaped, inverted U-shaped, S-shaped, W-shaped, or mixed relationships. These findings suggest that the benefits of internationalization may not be consistent and can vary depending on the stage or extent of a firm's internationalization.

When examining the linear relationship between internationalization and firm performance, there's a perspective suggesting that expanding business operations into

international markets will enhance a company's performance if it possesses sufficient conditions and competitive advantages to overcome the "cost of foreignness." Firstly, businesses can learn and accumulate experience from international markets, helping them gain market knowledge and identify risks in subsequent stages [7]. Secondly, international expansion also strengthens the company's position and relationships with suppliers and customers [15]. Furthermore, internationalization helps reduce operating costs and opens up opportunities to access new markets. Finally, businesses can diversify risks by operating in multiple markets, thereby avoiding institutional, political, and economic cycle risks [15].

Conversely, the internationalization process also comes with significant costs. For example, the legal responsibilities of being a "foreigner" can increase operating costs if the business is not part of a business network in that market [21]. Additionally, transportation costs and the cost of understanding foreign markets also increase, especially when the business lacks market knowledge [13]. At the same time, according to Contractor et al. [15], international operating costs and cultural distances can reduce the operational efficiency of businesses in the early stages of internationalization.

H1: Internationalization has a negative impact on the firm's performance.

2.5 Top manager's experience

The Uppsala model indicates that the managerial capacity of the leader plays a crucial role in promoting business growth in the international market [13, 14]. Upper Echelons Theory also suggests that a company's business results are influenced by the personal characteristics of the top manager [9]; internationalization strategy depends on the leader's personal perspective and strategic decisions. Accordingly, numerous empirical studies have affirmed the important role of the top manager's experience and gender [22] in shaping a company's internationalization behavior, further reinforcing the foundation of Upper Echelons Theory.

The top manager's accumulated knowledge over the years, developed from practical experience, becomes an essential factor in responding to the complexity and uncertainty of the international market. Specifically, this experience helps minimize "psychic distance"—the difference in culture, education, and perspectives between the company's country of origin and the international market [7]. Factors such as language differences, cultural disparities, and levels of economic development contribute to this distance, hindering information flow and increasing risks when businesses enter new markets. From internationalization experiences, managers will have more motivation and confidence to expand business operations into new markets [23].

Besides the expansion motivation as mentioned, the top manager's experience also has a direct impact on business results in the international market [24]. Hambrick and Mason [9] demonstrated that the psychological characteristics of the managers not only affect growth but also improve the company's business results. The manager's experience is considered a strategic resource that businesses can leverage to optimize benefits from international activities [25]. According to Sambharya [26], the benefits of a manager include the ability to reduce information costs, decrease anxiety, and enhance adaptability when a business enters foreign markets.

Furthermore, a manager's deep understanding of other countries' cultures and business practices enables the company to overcome cultural barriers by adapting products and branding strategies to each specific culture. This not only instills confidence in the manager's management but also contributes to risk reduction and profit

optimization, thereby enhancing the effectiveness of the company's international business operations. According to Crossland et al. [18], the top manager's extensive experience helps businesses build more effective strategies to navigate challenges in international markets.

H2: The relationship between internationalization and firm performance will be strengthened when the top manager has many years of experience.

2.6 Top manager's gender

Some Asian cultures tend to value masculine traits such as strength, power, independence, and competitiveness. Therefore, many studies have shown that businesses led by men tend to achieve more benefits than those led by women [27]. Typically, businesses founded or run by men have more stable financial resources, which is a crucial factor in internationalization, as this process requires high costs to build and maintain relationships and consolidate positions in foreign markets [28]. Moreover, male managers often face fewer barriers when entering international markets compared to women, creating favorable conditions for businesses in internationalization activities [29].

On the other hand, studies also indicate that gender diversity, especially in management roles, enhances firm performance by bringing diverse perspectives, improving decision-making processes, and fostering innovation. Concurrently, female managers possess the capacity to formulate creative strategic decisions, enhancing adaptability within the international environment [9]. In India, where the proportion of women in top management roles is higher than the global average, female managers can bring unique perspectives to optimize business strategies in the international environment, such as better cultural adaptability and relationship-building skills, supporting global expansion. Furthermore, the presence of women in top

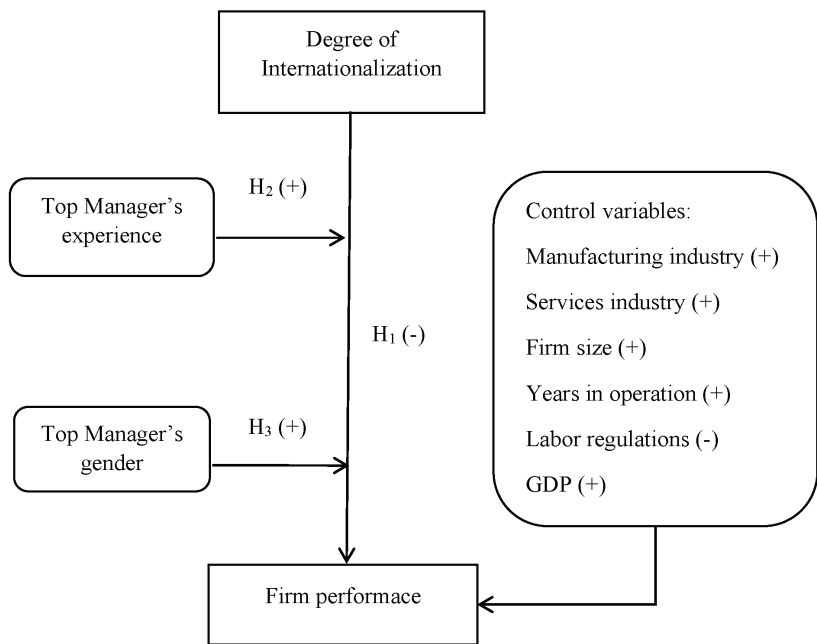


Figure 1.
Conceptual model. Source: Authors' compilation (2025).

management roles can also signal a progressive corporate culture that is more open to international collaborations and cross-border partnerships, further enhancing the benefits of internationalization for firm performance.

H3: The relationship between internationalization and firm performance will be strengthened when the top manager is a female.

From the literature review and theoretical basis, the author proposes the research model shown in **Figure 1**.

3. Methodology

3.1 Data

The analysis uses panel data from the World Bank Enterprise Surveys [30] covering 380 firms in India, representing both the manufacturing and service sectors. This dataset is widely used in international business research due to its standardized methodology, firm-level detail, and comparability across countries [31]. It allows for robust analysis of management characteristics and performance, particularly in emerging markets, and enhances the external validity of findings [5, 10].

3.2 Estimation method

This study employs Feasible Generalized Least Squares (FGLS) regression to examine the moderating effects of top management characteristics on the relationship between internationalization and firm performance. FGLS is chosen for its efficiency in correcting panel-level heteroscedasticity and autocorrelation, particularly under the Random Effects Model. The Hausman test indicated REM was appropriate (Prob > chi2 = 0.3019), while the Lagrange Multiplier test showed no significant heteroscedasticity (Prob > chibar2 = 1.0000). Multicollinearity was also not a concern as all VIF values remained below the 10.0 threshold [32]. The model specification is as follows:

$$ROS_{it} = \beta_0 + \beta_1 doi_{it} + \beta_2 expertm_{it} + \beta_3 gendertm_{it} + \beta_4 industry_se_{it} + \beta_5 industry_ma_{it} + \beta_6 firmsize_{it} + \beta_7 openyear_{it} + \beta_8 laborregulation_{it} + \beta_9 lngdp_{it} + \beta_{10} (doiex)_{it} + \beta_{11} (doige)_{it} + \beta_{it}$$

Where: ROS: dependent variable (firm performance); β_0 : intercept; β_1 - β_{11} : estimated coefficients of independent, moderating, and control variables; i : firm number; t : study period; ε_{it} : error term. The full regression results are reported in **Table 1**.

3.3 Variables and summary statistics

3.3.1 Dependent variable (firm performance) and independent variables

Several methods exist for measuring firm performance. Financial metrics like ROE [39], ROA [1], and ROS [40] are commonly used. This study used ROS due to its lower susceptibility to inflation and accounting principles, offering a more accurate performance reflection [41].

Construct	Measurement	Expectation
Dependent variable		
Firm performance (ros)	Returns on sales [33]	
Independent variable		
Degree of Internationalization (doi)	Overseas revenue over total revenue [33]	(+)
Moderating variable		
Top manager's experience (expertm)	Measured by the number of years of top manager's experience [34]	(+)
Top manager's gender (gendertm)	Top manager's gender is a dummy variable, getting value equal to 1 if the Top manager is Female and 0 if Male [35]	(+)
Control variables		
Services industry (industry_se)	Dummy variable (1: Services, 0: Others) [15]	(+)
Manufacturing industry (industry_ma)	Dummy variable (1: Manufacturing, 0: Others) [15]	(+)
Firm size (firmsize)	The number of employees can be measured by a dummy variable with 0 is SMEs (> = 5 to <=99 employees), 1 is large (> = 100 employees) [36]	(+)
Years in operation (openyear)	Year in operation as the number of years since the business has been in operation (year) [37]	(+)
Labor regulation (laborregulation)	Likert-5 levels: "To what extent, labor regulations are an obstacle to business operation." 0: no obstacles, 4: extreme obstacles [38]	(-)
GDP (lngdp)	GDP at purchaser's prices (World Bank)	(+)

Source: Authors' compilation (2025).

Table 1.
Explanation of variables in the research model.

Numerous studies employ various methods to measure internationalization. FSTS (Overseas revenue/Total revenue) is most common [1]. Ruigrok and Wagner [42] highlighted this as measuring the core financial aspect of internationalization. The FSTS calculation is:

$$\text{FSTS} = (\text{Overseas sales} / \text{Total sales}) \times 100(\%)$$

3.3.2 Moderating variable

Top manager's experience: is measured by the number of years of the top manager's experience [34].

Top manager's gender (gendertm): is a dummy variable, getting a value equal to 1 if the manager is Female and 0 if Male [35].

3.3.3 Control variable

Manufacturing industry: is represented as a dummy variable (1: Manufacturing, 0: Others) [15].

Other services industry: is represented as a dummy variable (1: Manufacturing, 0: Others) [15].

Firm size: this variable is measured by the number of employees and can be measured by a dummy variable with 0 is SMEs ($> = 5$ to ≤ 99 employees), 1 is large ($> = 100$ employees) [36].

Years in operation: measured in years since operation began [37].

Labor regulations: 5-point Likert scale (“To what extent are labor regulations an obstacle?” 0: No obstacle, 4: Extreme obstacle) [38].

GDP: measured at purchaser’s prices (World Bank).

4. Result

Table 2 presents descriptive statistics. The average ROS is 30.34%, ranging from -660% to $98,166\%$. The average degree of internationalization is 0.056% , ranging from $0-100\%$. **Table 3** presents the correlation matrix between variable pairs. **Table 4** presents the summary of regression results of the REM and FEM models. The moderated regression results with FGLS estimation are shown in **Table 5**.

The FGLS regression analysis supports all three hypotheses. Internationalization negatively affects firm performance ($\beta = -36.66$, $p < 0.001$), confirming H1. Managerial experience positively moderates this relationship ($\beta = 1.55$, $p < 0.001$), supporting H2. Female leadership also strengthens the internationalization-performance link ($\beta = 77.03$, $p < 0.001$), confirming H3.

5. Discussion

Model 3 in **Table 5** includes all variables in the research model. The value of Prob $> \chi^2_2 = 0.0000$ indicates the model’s fit with the actual data at a 99% confidence level.

According to Model 3, the relationship between the degree of internationalization and firm performance is a negative linear relationship, with the coefficient of the

Variables	Obs.	Mean	S.D	Min	Max
Firm performance (ROS) (%)	760	30.340	47.603	-660	98.166
Degree of internationalization (DOI) (%)	760	0.056	0.186	0	100
Top Manager’s experience (expertm)	760	20.369	12.647	1	70
Top Manager’s gender (gendertm)	760	0.049	0.215	0	1
Service industry (industry_se)	760	0.756	0.429	0	1
Manufacturing industry (industry_ma)	760	0.289	0.168	0	1
Firm size (firm size)	760	0.249	0.432	0	1
Years in operation (openyear)	760	30.819	15.432	4	92
Labor regulation (laborregulation)	760	1.005	1.078	0	4
GDP (lngdp)	760	28.6	0.260	28.34	28.86

Source: Authors’ compilation (2025).

Table 2.
Descriptive statistics of variables in the research model.

Variables	S.D	VIF	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1. ROS (%)	47603		1									
2. DOI (%)	0.186	4.43	-0.0449 ^{ns}	1								
3. Expertm	12.647	1.40	0.0098 ^{ns}	-0.0257 ^{ns}	1							
4. Gendertm	0.215	1.13	-0.0667 ^{ns}	-0.0096 ^{ns}	0.0519 ^{ns}	1						
5. Industry_se	0.429	1.40	0.0896 [*]	-0.0380 ^{ns}	0.0741 [*]	0.0143 ^{ns}	1					
6. Industry_ma	0.168	1.24	0.0237 ^{ns}	-0.0185 ^{ns}	0.0955 ^{**}	-0.0026 ^{ns}	-0.3044 ^{***}	1				
7. Firmsize	0.432	1.13	-0.0624 ^{ns}	0.1996 ^{***}	0.0764 [*]	0.0820 [*]	-0.1347 ^{***}	-0.0812 [*]	1			
8. Openyear	15.432	1.27	-0.0033 ^{ns}	-0.0123 ^{ns}	0.3434 ^{***}	0.0387 ^{ns}	0.0005 ^{ns}	-0.0402 ^{ns}	0.1390 ^{***}	1		
9. Laborregulation	1.078	1.06	-0.0300 ^{ns}	-0.0731 [*]	-0.1281 ^{***}	0.0102 ^{ns}	-0.0428 ^{ns}	-0.0300 ^{ns}	-0.0198 ^{ns}	0.1172 ^{**}	1	
10. Lngdp	0.260	1.50	0.1670 ^{**}	-0.1331 ^{***}	0.2448 ^{***}	-0.0183 ^{ns}	0.3649 ^{***}	0.1727 ^{***}	-0.2040 ^{***}	-0.1549 ^{***}	-0.1295 ^{***}	1

Note: (***), $p < 0.001$; (**), $p < 0.01$; (*), $p < 0.05$; (ns) $p > 0.10$ (ns = non significance); values in parentheses are standard errors. Source: Authors' compilation (2025).

Table 3.
 Descriptive statistics and correlation matrix of variables in the model ($n = 760$).

Variables	Firm performance: ROS	
	REM	FEM
Constant	−861.567 ^{***} (226.741)	−517.616 ^{ns} (274.851)
<i>Control variables</i>		
Industry_se	3.741 ^{ns} (4.674)	13.407 ^{ns} (6.967)
Industry_ma	3.464 ^{ns} (11.269)	9.259 ^{ns} (16.280)
Firmsize	−2.631 ^{ns} (4.167)	−10.161 ^{ns} (7.901)
Openyear	0.160 ^{ns} (0.124)	0.187 ^{ns} (0.219)
Laborregulation	−0.689 ^{ns} (1.622)	−4.769 [*] (2.310)
Lngdp	31.204 ^{***} (7.970)	19.036 [*] (9.667)
<i>Main effect</i>		
Doi	−51.820 ^{**} (19.229)	−41.413 ^{ns} (28.051)
Expertm	−0.291 ^{ns} (0.158)	−0.189 ^{ns} (0.226)
Gendertm	18.761 [*] (8.376)	−25.767 [*] (12.007)
<i>Interaction terms</i>		
Doiex	92.077 ^{ns} (57.332)	1.666 ^{ns} (1.257)
Doige	2.298 ^{**} (0.885)	116.628 ^{ns} (82.082)
Observations	760	760
	Wald chi2 (11) = 39,87	F(11.369) = 3.60
	Prob> chi2 = 0.0000	Prob>F = 0.0001
	R ² within = 0.0801	R ² within = 0.0969

Note: (***), $p < 0.001$, (**), $p < 0.01$, (*), $p < 0.05$, (ns), $p > 0.1$ (non significance), values in parentheses are standard errors. Source: Authors' compilation (2025).

Table 4.
REM and FEM models.

Variables	Firm performance: ROS		
	Model 1	Model 2	Model 3
Constant	−766.656 ^{***} (34.297)	−831.818 ^{***} (37.976)	−873.023 ^{***} (41.502)
<i>Control variables</i>			
Industry_se	2.423 ^{**} (0.851)	3.140 ^{***} (0.865)	3.303 ^{***} (0.913)
Industry_ma	−0.912 ^{ns} (1.905)	0.739 ^{ns} (2.145)	1.248 ^{ns} (2.611)
Firmsize	−3.179 ^{***} (0.644)	−2.438 ^{***} (0.660)	−3.016 ^{***} (0.736)
Openyear	0.881 ^{***} (0.019)	0.133 ^{***} (0.023)	0.166 ^{***} (0.022)
Laborregulation	−0.710 ^{**} (0.245)	−0.927 ^{**} (0.268)	−0.696 [*] (0.281)
Lngdp	27.760 ^{***} (1.208)	30.091 ^{***} (1.336)	31.577 ^{***} (1.465)
<i>Main effect</i>			
Doi		−5.393 ^{***} (1.297)	−36.664 ^{***} (5.460)
Expertm		−0.149 ^{***} (0.031)	−0.266 ^{***} (0.0341)

Variables	Firm performance: ROS		
	Model 1	Model 2	Model 3
Gendertm		−12.193*** (1.338)	−12.434*** (1.911)
<i>Interaction terms</i>			
Doiex			1.546*** (0.238)
Doige			77.032*** (19.876)
Observations	760	760	760
Wald chi2	834.08***	1297.24***	980.23***
<i>Note: (***) $p < 0.001$, (**) $p < 0.01$, (*) $p < 0.05$, (ns) $p > 0.10$ (non significance), values in parentheses are standard errors. Source: Authors' compilation (2025).</i>			

Table 5.
Regression model results using the Feasible Generalized Least Squares (FGLS) estimation method.

variable DOI being negative (36.664), thus supporting hypothesis H1. Businesses new to international operations often face substantial initial costs in exploring and penetrating foreign markets. These costs include investments to adapt products to international market requirements, which do not arise if the business operates only domestically [7]. Furthermore, in the early stages of internationalization, as businesses expand into new markets, they face liabilities of foreignness, especially if they lack an established business network in that market [21]. Along with that, businesses also encounter liabilities of newness when entering the international market, including finding human resources, capital, and customers. Additionally, the geographical distance between international and domestic markets increases transportation costs for businesses [15]. Because business managers often lack knowledge and information about foreign markets, they may incur additional costs to learn and adapt to the business environment in new countries [13].

Model 3 in **Table 5** indicates a positive correlation between the interaction of the degree of internationalization and the managerial experience of the leader. With the interaction coefficient doiex being positive (1.55) and statistically significant ($p < 0.001$), this result supports the initial expectation and accepts hypothesis H2. This implies that the relationship between internationalization and firm performance is strengthened when top managers have more managerial experience. Increased years of experience often equate to accumulated knowledge and skills, creating momentum and a foundation for expanding into international markets [7]. The top manager's experience provides support, helping businesses overcome barriers and challenges in the international environment, such as cultural differences, legal regulations, and supply chain management complexities. According to Carpenter and Fredrickson [43], experienced managers tend to make faster and more effective decisions in complex international environments due to their extensive knowledge and ability to adapt to various situations. Particularly in the Indian context, where cultural and regulatory barriers can be more complex due to significant differences between domestic and international markets, managerial experience acts as a "lever" to help businesses achieve higher operating efficiency. The study by Elango and Pattnaik [44] also emphasizes that the international experience of managers enhances the competitiveness of businesses when expanding into new markets (**Figure 2**).

In Model 3 of **Table 5**, the interaction between the level of internationalization and the manager's gender reveals a positive correlation supporting Hypothesis H3

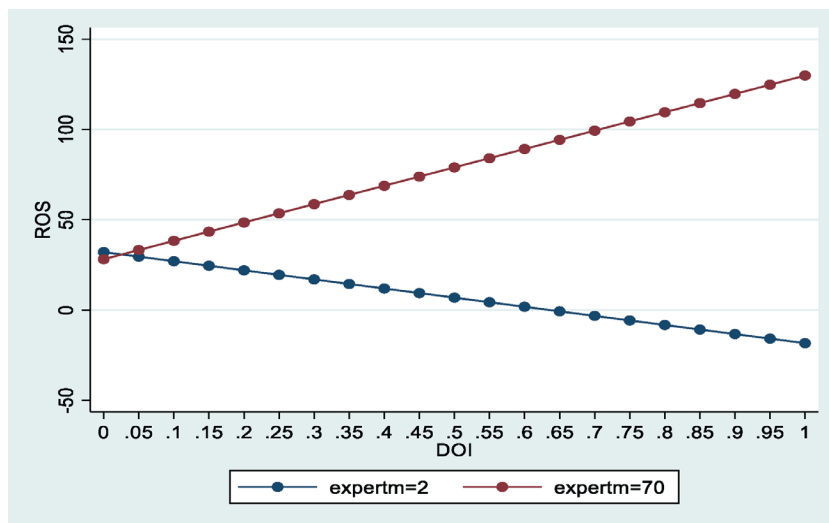


Figure 2.

Moderating effect of top manager's experience on internationalization and firm performance. Source: Authors' compilation (2025).

(positive interaction coefficient of 77.032 and statistically significant ($p < 0.001$)). This finding suggests that the relationship between the level of internationalization and firm performance is strengthened when the business has a female top manager. Conversely, when the manager is male, the positive impact of internationalization on firm performance tends to decrease. Female managers are often recognized for their flexible management skills, empathy, and multidimensional strategic thinking—qualities that support businesses in navigating the diverse and complex international market. Dezső and Ross [20] indicate that women in top management positions tend to promote organizational innovation through their ability to diversify perspectives and foster creativity. In the context of internationalization, where businesses face cultural diversity and product adaptation requirements, this becomes particularly important. Specifically, research by Post and Byron [19] also shows that female managers excel in adapting to different cultural factors, building relationships, and managing conflicts. These skills help businesses enhance their adaptability in a rapidly changing international environment, thereby improving firm performance (**Figure 3**).

Additionally, the remaining variables in the model also influence firm performance (ROS), including: services industry ($\beta = 3.303$; $p < 0.000$), years in operation ($\beta = 0.166$; $p < 0.000$), and GDP ($\beta = 31.577$; $p < 0.000$) having a positive impact on ROS; top manager's experience ($\beta = -0.266$; $p < 0.000$), top manager's gender ($\beta = -12.434$; $p < 0.000$), firm size ($\beta = -3.016$; $p < 0.000$), and labor regulation ($\beta = -0.696$; $p < 0.013$) having a negative impact on ROS. The remaining variable, manufacturing industry, is not statistically significant.

5.1 Research limitations and future research directions

This study has some limitations: (1) it only considers two personal moderating factors of top manager's experience and top manager's gender; influencing the relationship between internationalization and firm performance; (2) the sample is

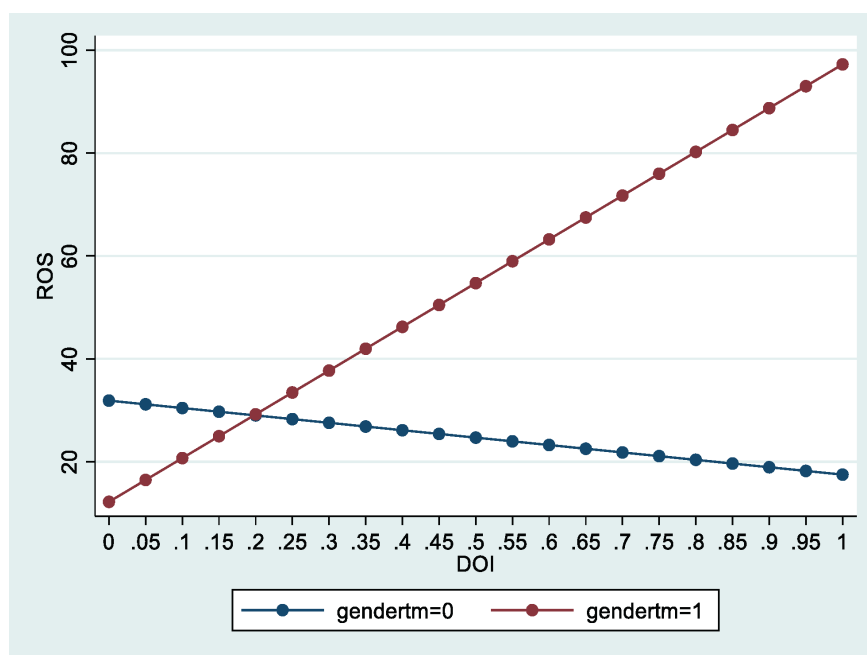


Figure 3.
 Moderating effect of top manager's gender on internationalization and firm performance. Source: Authors' compilation (2025).

limited to a single country (India); and (3) the degree of internationalization and firm performance are measured using single indicators.

Based on these limitations, suggestions for future research include: (1) expanding the analysis to include other moderating factors such as cultural, geographical, and economic distances and the characteristics of board members; (2) broadening the scope to multiple countries and analyzing different stages of international market entry; and (3) employing multidimensional measurement methods to provide a more comprehensive understanding of the relationship between internationalization and firm performance.

5.2 Implications

5.2.1 Theoretical implications

This research contributes to the Uppsala model and the Upper Echelons Theory by illustrating how top managers' attributes influence the relationship between internationalization and firm performance, especially in developing economies.

The Uppsala model [7, 8] posits that firms internationalize incrementally through experiential knowledge acquisition, mitigating risks tied to foreign market entry. This study expands the model by showing how top managers' experience and gender can affect this process. Managerial experience enhances firms' ability to navigate international markets by expediting learning, bridging information gaps, and facilitating more effective strategic decision-making. This challenges the traditional view of internationalization as an exclusively organizational process and underscores the critical role of individual decision-makers. Furthermore, female top managers can positively influence firm performance during internationalization due to their

cross-cultural management adaptability and relationship-building skills. This suggests that the Uppsala model should integrate individual managerial competencies with firm-level learning to better assess the speed and success of internationalization.

This study supports the Upper Echelons Theory [9] by demonstrating that top managers' characteristics significantly impact strategic decisions and firm performance during internationalization. Experienced managers can mitigate the liabilities of foreignness using their knowledge, networks, and foresight, leading to more effective strategies. This aligns with the Upper Echelons perspective that managerial cognition and experience shape organizational outcomes, especially in uncertain environments. Furthermore, leadership gender diversity improves internationalization outcomes, as female top managers often exhibit greater flexibility, stronger cross-cultural competencies, and superior relationship management—crucial for success in foreign markets. This contributes to the discussion on executive diversity, suggesting that gender-diverse leadership teams may achieve better performance globally.

5.3 Practical implications

For management and policy, this study offers key insights into how firms in developing economies can improve internationalization strategies by leveraging top managers' characteristics. Firms should not rely exclusively on the sequential approach of the traditional Uppsala model but should also account for managerial experience and leadership diversity. Investing in leadership development that enhances international exposure and cross-cultural competencies can significantly improve firms' ability to operate abroad. Prioritizing gender diversity in executive roles is also beneficial, as female managers demonstrate strategic advantages in international business. Policymakers should support firms with targeted training and resources that facilitate knowledge acquisition and leadership development for managers in early internationalization stages. These measures will help firms overcome initial obstacles and optimize global market performance.

5.4 Recommendations

This study presents key managerial implications for boards and policymakers in developing economies. First, boards should prioritize recruiting experienced female managers, as female participation in top management not only improves adaptability but also benefits internationalization, contributing to enhanced performance in the global competitive landscape. Simultaneously, businesses should develop phased internationalization strategies. Initially, firms face high costs and risks, potentially hindering performance, but once established internationally, the benefits of internationalization increase significantly, improving performance in later stages. Finally, managers must recognize the role of personal characteristics, such as management experience, in shaping internationalization strategies. To support these objectives, training programs for female managers and policies supporting small and medium-sized enterprises during initial internationalization should be prioritized, fostering sustainable and effective development in the international environment.

6. Conclusion

This study employs Feasible Generalized Least Squares (FGLS) regression analysis to identify factors influencing the relationship between internationalization and firm

performance in a developing economy, with India as the focus. Results indicate a negative linear relationship, with the top manager's experience and the fact that the top manager is female positively moderating this relationship. Specifically, Hypothesis H1 is supported, consistent with Transaction Cost Theory [15], which emphasizes that high initial internationalization costs can reduce firm performance. Furthermore, Hypothesis H2 is supported, indicating that the top manager's experience possesses a greater understanding of international markets, reducing information costs and risks [18]. This contrasts with Phan Anh Tu et al. [45] but aligns with Tu and Thanh [46], which examined top management characteristics in a transitional economy, and with Hsu et al. [10], which focused on CEO experience. Finally, Hypothesis H3 is supported, suggesting female managers enhance firm performance through innovative thinking and flexible management, particularly during internationalization [9]. This aligns with Tu and Thanh [46]. Therefore, understanding performance differences in internationalizing firms, especially in developing economies, requires considering top management characteristics.

The study also analyzes control variables and factors directly influencing operational effectiveness. The variables services industry, years in operation, and GDP show a positive correlation with firm performance, while the top manager's experience, the top manager's gender, firm size, labor regulation, and degree of internationalization exhibit an inverse relationship. Notably, the manufacturing industry variable lacks statistical significance but is included in the model due to its contextual importance in business.

This study extends internationalization theory by integrating the top management team's individual moderating factors. By combining the Uppsala theory [7, 13, 14] and Upper Echelons Theory [9], the study provides a comprehensive theoretical framework to explain how top manager characteristics influence internationalization strategy.

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